

Root Cause Analysis

Asked by : Google



Problem Statement : You are the product manager at Google and you see there is 10% drop in daily active users of Google Maps. Find the root cause of this ?

Clarifying Questions :

Q1 : What do we mean by Daily Active Users, for example individual who are just want navigation to a place or individuals who upload a review etc.

Ans : Individuals who are using Google Maps for navigation.

Q2 : Does this also counts companies who have integrated google maps for navigation for example ride sharing apps like Rapido, Uber etc. ?

Ans : No, there is a decline in individuals who comes to GMaps for navigation.

Q3 : Since when we are seeing this decline ?

Ans : Its been a week.

Q4 : Is this impact across a specific place (country, city etc.) or its globally ?

Ans : Its globally

Q5 : Is this specific to website or an app ?

Ans : This decline is across both

Q6 : Are we seeing this drop across a specific demography of people ?

Ans : No, its across all the demographics.

Q7 : Are we seeing this drop across a specific demography of people ?

Ans : No, its across all the demographics.

Q8 : Whether this is an industry wide problem or not ?

Ans : No, its not across other video sharing apps.

Sanity Check :

Q1 : Have we changed the definition of how the metric is calculated ?

Ans : No we haven't changed any definition.

Q2 : Is our metric tool working fine ?

Ans : Yes it is.

Lets focus on external factors for now, to find out the reason for the impact.

External Factors :

Q1 : Is there any rise of a new competitor ?

Ans : No new competitor has entered the market.

Q2 : Have the existing competitor launched any new features ?

Ans : No

Q3 : Has there been any good or bad PR against us ?

Ans : No, nothing as such.

Q4 : Have we violated any regulations or promoted something which we shouldn't ?

Ans : No, we have not violated any regulations.

Q5 : Is this a season drop in the sense people like to stay at home for vacations and do not want to travel out ?

Ans : Nothing as such

Q6 : Is there any environmental disaster because of which people are staying at home ?

Ans : No there isn't.

Since there are no external factors which are impacting this decline, lets focus on Internal Factors.

Internal Factors :

Q1 : Is there any issue with accuracy of GMaps navigation ?

Ans : No there isn't

Q2 : Have we deployed any new feature ?

Ans : No we haven't deployed anything.

Q3 : Is there any change in UI/UX ?

Ans : There is no change in the user journey.

Q4 : Are there any technical bugs which could have caused this decline ?

Ans : As of now there are no bugs.

Q5 : Are we facing any latency issues like problem in loading ?

Ans : Yes we are.

Q6 : Are we facing latency issues in search ?

Ans : Yes we are.

I would go into greater depth of this while discussing the customer journey.

Customer Journey :

New User Journey

Q1 : Has the user faced any delay in response while opening the app or the website ?

Ans : No

Q2 : Is there a decline in %age of (# of users who clicks on search option/ # of users who comes to GMaps) ?

Ans : No there is no decline.

Q3 : Does the user face delay in loading of the searched result ?

Ans : Yes there is.

Q4 : Is there a increase in %age of of users who dropped off immediately after searching ?

Ans : Yes there is a significant increase in that.

Q5 : Is there a increase in %age of users dropping off after getting the search results ?

Ans : No there is no decline

According to the customer journey proves there is increase in %age of users who drop off immediately after searching and there is no drop off in any other steps, we also discussed about the latency issue while performing a search.

Hence the delay in getting search results has caused a decline of 10% in DAUs.

Interviewer : What could have caused this issue ?

Ans : This could be because of -

1. Search index of the maps is not working properly leading to delayed searches.
2. Images in the maps could be of high resolution which could have resulted more load.